MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PERIODIC TEST – 04 (POLYNOMIALS)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 2: Polynomials** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided in Section D.
2. **Section A** comprises 5 questions (4 MCQs and 1 Assertion-Reason) of 1 mark each.
3. **Section B** comprises 3 VSAQ questions of 2 marks each.
4. **Section C** comprises 3 SA questions of 3 marks each.
5. **Section D** comprises 2 Long Answer / HOTS questions of 5 marks each.

SECTION A: Objective & MCQs (5 × 1 = 5 Marks)

Q1. If α and β are zeroes of $x^2 - 2x - 15$, then the polynomial whose zeroes are 2α and 2β is: [1]

(A) $x^2 - 4x - 60$ (B) $x^2 + 4x - 60$ (C) $x^2 - 4x + 60$ (D) $x^2 - 8x - 30$

Q2. If α, β are zeroes of $x^2 - 3x - 2$, then the value of $\alpha^3 + \beta^3$ is: [1]

(A) 45 (B) 27 (C) 36 (D) 18

Q3. If the zeroes of $f(x) = 2x^2 + 5x - 12$ are α and β, then $\alpha\beta^2 + \alpha^2\beta$ is equal to: [1]

(A) 15 (B) -15 (C) 30 (D) -30

Q4. If α, β are the zeroes of $f(x) = x^2 - 4$, then $\alpha^4 + \beta^4$ is: [1]

(A) 32 (B) 16 (C) 64 (D) 0

Q5. **Assertion (A):** The polynomial $f(x) = x^2 - 2x - 3$ has zeroes 3 and -1.
Reason (R): Zeroes of a polynomial $p(x)$ are the x-coordinates of the points where $y = p(x)$ crosses the x-axis. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Questions (3 × 2 = 6 Marks)

Q6. If the sum of the zeroes of the quadratic polynomial $f(x) = kx^2 - 3x + 5$ is 1, find the value of k. [2]

Q7. If one zero of the polynomial $f(x) = 3x^2 + 8x + k$ is reciprocal of the other, find the value of k. [2]

Q8. Find the zeroes of $p(t) = 5t^2 - 4 - 8t$ and verify the relationship between zeroes and coefficients. [2]

SECTION C: Short Answer Questions (3 × 3 = 9 Marks)

Q9. If α and β are zeroes of the quadratic polynomial $f(x) = x^2 - x - 4$, find the value of $\frac{1}{\alpha} + \frac{1}{\beta} - \alpha\beta$. [3]

Q10. If α and β are the zeroes of the quadratic polynomial $f(x) = 3x^2 - 4x + 1$, find a quadratic polynomial whose zeroes are $\frac{\alpha^2}{\beta}$ and $\frac{\beta^2}{\alpha}$. [3]

Q11. If α, β are zeroes of the polynomial $x^2 - 6x + a$ and $3\alpha + 2\beta = 16$, find the value of a. [3]

SECTION D: Long Answer / HOTS Questions (2 × 5 = 10 Marks)

Q12. If α and β are the zeroes of $p(x) = 3x^2 - 4x + 1$, form a quadratic polynomial whose zeroes are $\frac{\alpha^2}{\beta}$ and $\frac{\beta^2}{\alpha}$. Also, evaluate the expression $\alpha^4\beta^2 + \alpha^2\beta^4$. [5]

Q13. A quadratic polynomial $f(x) = ax^2 + bx + c$ has zeroes α and β. [5]

(i) Prove that $\frac{1}{\alpha} + \frac{1}{\beta} = -\frac{b}{c}$.

(ii) If $c = a$, prove that the zeroes are reciprocals of each other.

(iii) If $b = 0$, prove that the zeroes are equal in magnitude and opposite in sign.

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